

ZHEYUAN CHEN

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EDUCATION

University of California, Santa Cruz
Ph.D., Computer Science and Engineering

Sept 2024 – Present

RESEARCH & WORK EXPERIENCE

Microsoft Research

Research Intern, RiSE Group

Jun 2025 – Sep 2025

Redmond, WA

- Developed **SIMT-Step**, a formal operational semantics for GPU warp execution, providing a rigorous model of divergence, reconvergence, and subgroup behavior across architectures.
- Built a Clang AST-based interpreter and fuzzing framework for HLSL to automatically generate warp-execution tests.
- Discovered and reported numerous warp-execution bugs under non-uniform control flow across NVIDIA, Intel, and AMD GPUs, revealing inconsistencies in hardware and compiler behavior.

Mercedes-Benz Research & Development North America

Software Engineering Intern

Jun 2024 – Dec 2024

Sunnyvale, CA

- Contributed to middleware for automated driving systems (ADS), improving communication, data flow, and resource sharing across safety-critical components.
- Developed an automated toolchain to migrate a large Bazel project (CUDA/C++) to SYCL, reducing manual effort by 90% and improving portability across heterogeneous hardware.

Languages, Systems, and Data (LSD) Lab

Research Assistant

Jan 2023 – Present

University of California, Santa Cruz

- Developed a formal operational semantics for GPU warp execution to address the lack of rigorous semantic models in existing GPU programming languages, and verified key behaviors using TLA+.
- Helped launch the **WebGPU** backend in llama.cpp, with primary contributions to the portable **FlashAttention** kernel in **WGSL**.

PUBLICATIONS

Zheyuan Chen, Naomi Rehman, Guido Martínez, and Tyler Sorensen. “SIMT-Step Execution: A Flexible Operational Semantics for GPU Subgroup Behavior.” *PLDI 2026* (accepted).

Yanwen Xu, Rithik Sharma, **Zheyuan Chen**, Shaan Mistry, and Tyler Sorensen. “BetterTogether: An Interference-Aware Framework for Fine-grained Software Pipelining on Heterogeneous SoCs.” *IISWC 2025*. **Best Paper Award**.

SKILLS

Languages: Rust, C/C++, Python, Go, HLSL, GLSL, Metal, SPIR-V, TLA+

Frameworks & Systems: CUDA, SYCL, Vulkan, HIP, OpenCL, LLVM, MLIR